

Task	Task Description	Person	Estimated hours for the task	Actual hours used
Github Readme	Writing the readme document with a description of the engine, how to use the engine, controls and each specific action or option, extra engine features, additional comments.	Pablo	2	3
StreetEnvironment	The 'StreetEnvironment.fbx' scene is automatically loaded at startup. Texture adjustments, positions, parent and childs, and cleanup of corrupted names.	Pablo	3	3
GameObjects	Hierarchy: delete, reparent, create empty, and create children. - Transform: translation, rotation, and scaling of Game Objects. - Mesh: select or drag and drop any imported mesh. - Texture: select or drag and drop any imported texture. - The camera is a component with configurable settings.	Pablo	9	10
Custom Format	Models, meshes, and textures are saved in their own format in the Library folder.	Pablo	8	8
Scene Serialization	The scene can be serialized to a file that can be loaded.	Pablo	9	7
ImGuizmo	Added ImGuizmo for easier engine use.	Pablo	1	1
Grid	Added a grid to the scene for better orientation in the engine scene.	Pablo	1	1
Bug Fixing	ID conflict on GameObjects with ImGui in delivery 1 Windows configuration and display The Engine Console was not displaying all the engine data. Improved the Logs system to add more transparency to code processes. Bug F and Alt in delivery 1 Camera Component Windows Dialog usage Transparency Textures	Pablo	10	12

Github Readme	Finish developing the frustum culling and mouse picking sections of the readme. And add an explanation of the Octree window.	Víctor	1	1
Mouse Picking	Implementation of Raycasting from the camera to be able to select GameObjects directly in the 3D scene by clicking with the mouse.	Víctor	4	6
Frustum culling	Generation of AABBs (collision boxes) for all meshes. Logic to discard objects outside the camera's frustum from rendering. Debug display of boxes and raycasts.	Víctor	6	12
Octree	Desarrollo de la estructura de datos Octree para particionar el espacio. Integración del Octree para acelerar las consultas de Mouse Picking y Frustum Culling, evitando iterar sobre todos los objetos.	Víctor	8	9
Bug fixing	Troubleshooting ray intersection issues, incorrect AABB display, and optimising object insertion into the Octree.	Víctor	2	4
Time Management	Implementation of simulation controls: Play, Pause, and Stop buttons. Logic for saving the initial state of the scene when pressing Play and restoring it when pressing Stop. Timer management (Game Time vs Real Time).	Claudia	6	12
Resource Management	Creation of the 'Assets' window to view files. Implementation of Drag & Drop to import files. Deletion synchronisation between Assets and Library folders. Reference counting system to avoid duplicates in memory. Generation of .meta files.	Claudia	14	22
Bug fixing	Fixed bugs related to resource loading, duplicate texture imports, and delta time issues when	Claudia	4	4

	pausing the simulation. Fixed critical crash on Play/Pause transition when modifying hierarchy (creating/moving objects).			
Delivery	Preparation of files, executable, task list, and delivery.	All	2	2